iNUX Interactive Documents: **Best Practice Examples** for Implementation in Education (Basic Hydrogeology)

04. February 2026; v1.1

by Thomas Reimann[1], Rudolf Liedl[1], Markus Giese[2], Roland Barthel[2], Edith Grießer[3], Steffen Birk[3], Oriol Bertran[4], Daniel Fernàndez-Garcia[4]

[1]*TU Dresden, Institute for Groundwater Management*

[2]*University of Gothenburg, Department of Earth Sciences*

[3]*University of Graz, Department of Earth Sciences, NAWI Graz Geocenter*

[4]*Polytechnical University of Catalonia, Department of Civil and Enviro. Eng.*

Disclaimer

This document represents a static snapshot of the *iNUX Best Practice Examples for Implementation in Education* at the time of publication.

The most recent online version is available at the gw-inux GitHub repository:

https://github.com/gw-inux/iNUX-Handbook/tree/main/Best_Practice

Content

Purpose and Scope

Goal of the report

To document how interactive digital tools were implemented in hydrogeology education (Basic Hydrogeology), using concrete examples that illustrate transferable best practices. The focus is on didactic integration, practical effort, and lessons learned, rather than on technical details of the tools themselves.

Target audience

University teachers, course coordinators, and educational developers interested in adopting interactive tools in lectures, exercises, or complete courses.

Best practice examples for Basic Hydrogeology

The example presented in this report represent a selected subset of best practices and is intended to illustrate typical use cases and transferable practices. A more comprehensive and continuously updated collection of interactive tools, documentation, and accompanying resources is available online via the project's web platform under *https://github.com/gw-inux/iNUX-Handbook/tree/main/Best_Practice*.

Best Practice Example – Aquifer Response on Water Abstraction

This example presents a best-practice implementation of an interactive visualization in an undergraduate hydrogeology course, aimed at improving conceptual understanding of pumping-induced drawdown in confined and unconfined aquifers.

1. Short Description of the Interactive Educational Tool

The interactive tool 'Flow2Well_transient_confined_unconfined.py' (available in the iNUX interactive materials with index 03-05-008 and online under https://flow2well-transient-unconfined-confined.streamlit.app/) visualizes transient groundwater flow to a pumping well under confined and unconfined aquifer conditions. Based on the Theis solution, it allows direct, side-by-side comparison of drawdown evolution by interactively changing pumping rate and aquifer parameters, Figure 1. The tool is designed for live education in lectures and classroom settings. Immediate visual feedback on parameter changes supports intuitive understanding of key differences in aquifer response without requiring prior modeling experience.

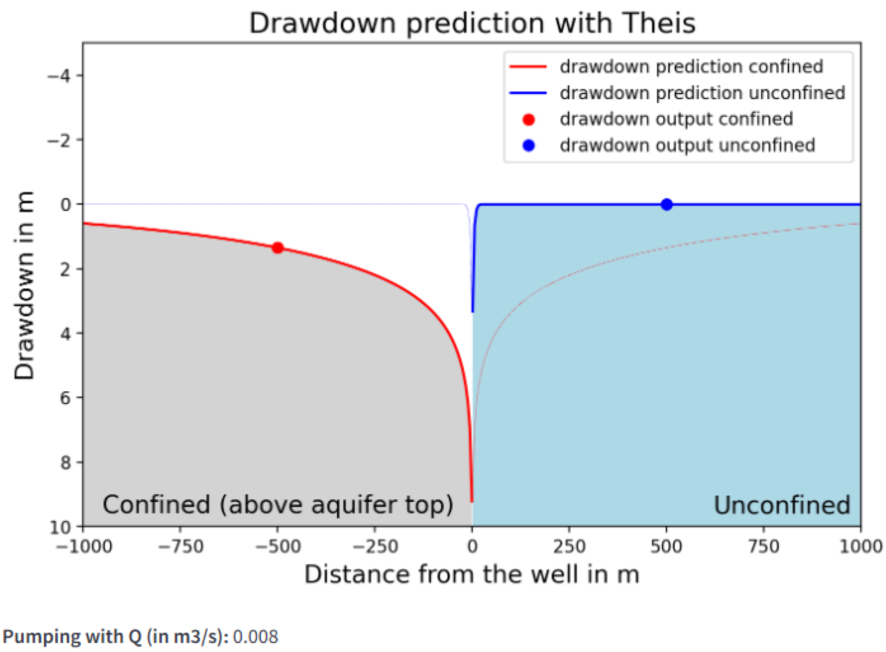

Figure 1: The interactive Streamlit app with the radial distribution of drawdown in response to water abstraction.

2. Context and educational setting

The interactive tool was implemented in the undergraduate course GV2002 *Hydrology and Hydrogeology* at the University of Gothenburg in autumn 2024. The course addresses fundamental hydrological and hydrogeological concepts and is attended by students from the Earth Sciences Department (Geology). Students typically have a background in Earth sciences with a stronger emphasis on geological interpretation than on advanced mathematics or physics, and therefore benefit from a qualitative, concept-driven approach to hydrogeological processes.

The application was used during a regular classroom lecture as a live, instructor-guided visualization. The primary role of the app was conceptual illustration, not numerical problem solving. The tool complemented traditional slides and board explanations and was not used as a standalone self-learning resource. The class size was moderate, with around 20 students (a typical undergraduate cohort), allowing for short, informal, interactive exchanges between the instructor and students.

3. Learning objective(s)

Conceptual understanding

- Explain the fundamental conceptual differences between confined and unconfined aquifers, particularly with respect to storage mechanisms and their influence on transient drawdown behavior.

Process interpretation

- Interpret how groundwater drawdown develops in space and time in response to pumping and describe how this evolution differs between confined and unconfined aquifers.

Parameter sensitivity and comparison

- Identify the key parameters controlling drawdown (e.g. transmissivity, storage coefficient or storativity) and explain how changes in storage properties can make confined and unconfined aquifer responses more comparable, thereby clarifying their conceptual relationship.

4. Tool integration and didactic design

The interactive tool was implemented during a classroom lecture within a teaching unit on hydrogeological properties of rocks, soils, and geological formations. At this stage of the course, students were familiar with key aquifer properties such as transmissivity and storativity, but had not yet been formally introduced to flow-to-well concepts.

The teaching sequence started with a short classroom assessment that activated prior knowledge and elicited students' initial understanding, see Figure 2. Questions addressed hydrogeological properties (including transmissivity and storativity) and asked students to qualitatively predict how drawdown propagates from a pumping well over space and time.

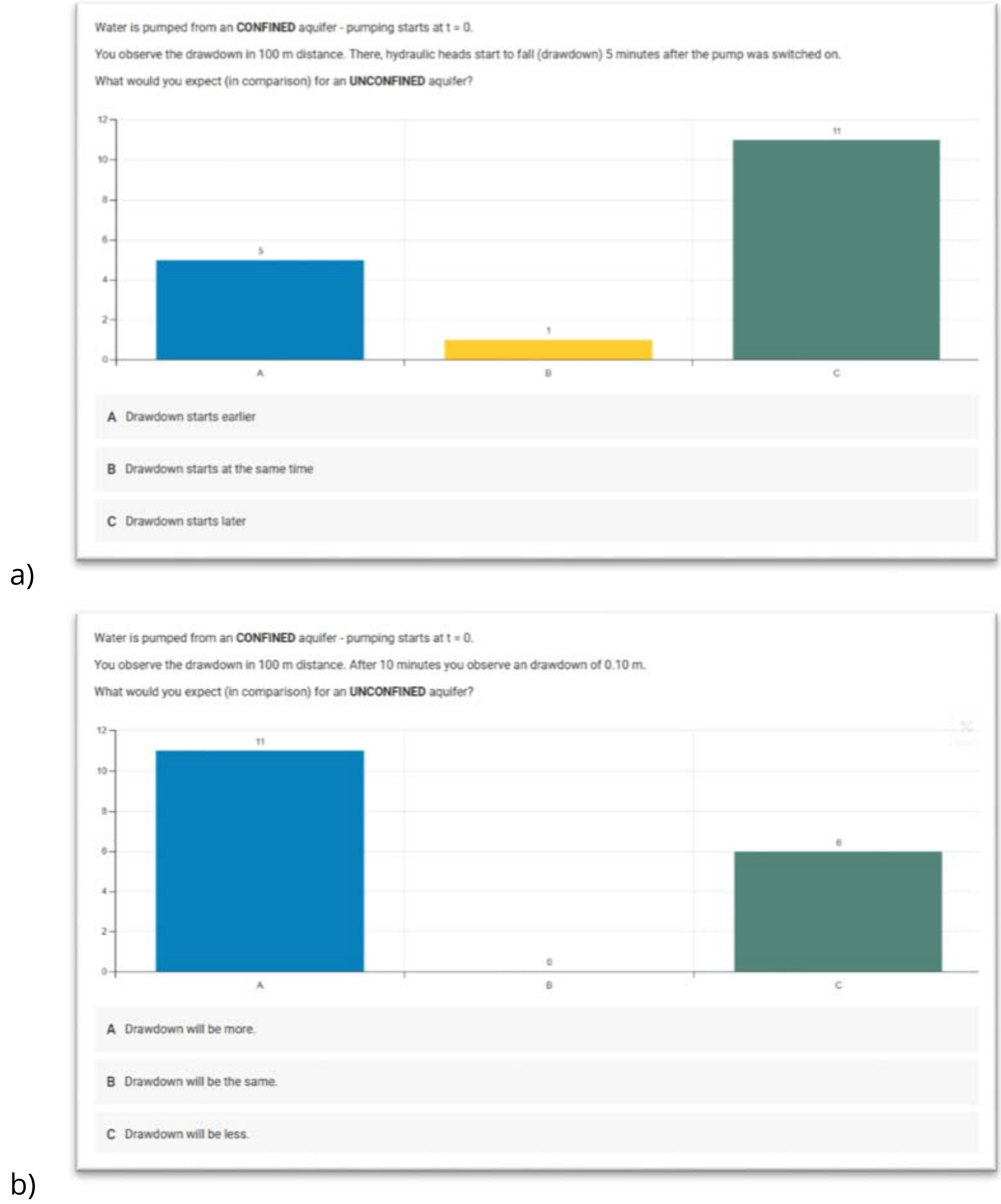

Figure 2: Two questions from the classroom assessment with the answers, reflecting the students' initial understanding.

Following this assessment, the interactive tool was introduced and briefly explained. The instructor outlined the conceptual setup (central pumping well, adjustable abstraction rate, and aquifer properties) and demonstrated how drawdown evolves in both confined and unconfined aquifers. The live interaction with the tool lasted approximately 10–15 minutes and was fully instructor-guided. Students were encouraged to follow the instructions with their own devices, which was possible due to the platform-independent tool that was provided through the Streamlit SharedCloud.

During the demonstration, particular emphasis was placed on comparing confined and unconfined responses under otherwise identical conditions. Because transmissivity was kept constant in both cases, the observed differences in drawdown behavior could be clearly attributed to storage properties. Through guided discussion, students identified that the much smaller storativity of confined aquifers is the key reason for the differing system responses.

To deepen this insight, the storativity of the unconfined aquifer (represented by specific yield) was systematically reduced. Students observed that, as storativity decreased, the unconfined aquifer response increasingly resembled that of a confined system, accompanied by a reduction in drawdown. This interactive exploration provided a concrete conceptual bridge between aquifer type, parameter range, and system behavior.

The session concluded with conventional lecture slides that formalized the concepts of storage properties and storativity, explicitly linking the theoretical explanations to the observations made during the interactive demonstration.

5. Implementation effort and technical setup

The required technical infrastructure for this implementation was minimal. The interactive tool is publicly available via Streamlit Cloud and can be accessed through a standard web browser without installation or login.

Classroom assessment questions can be implemented using common audience response systems (e.g., Partici.fi) or asked directly in class as verbal questions. Preparation effort was low and mainly consisted of adding a single slide containing QR codes linking to the classroom assessment and to the Streamlit application.

Once prepared, the approach is highly reusable. The same didactic structure can be applied to related topics such as flow-to-well concepts or pumping test interpretation with little or no modification.

The required instructor expertise is intermediate. While the tool itself is easy to operate, the interactive setting may trigger follow-up questions from students that require a solid conceptual understanding of groundwater flow and storage processes.

6. Observed outcomes and feedback

During the classroom implementation, students followed the demonstration attentively, with some additionally exploring the tool on their own devices. The interactive adjustment of storage parameters allowed a direct visual comparison between confined and unconfined aquifer responses, making differences in system behavior easy to illustrate.

The time-dependent evolution of drawdown could be demonstrated effectively using the time slider, supporting discussion of how drawdown propagates during ongoing pumping. This aspect is often difficult to convey using static figures alone.

After the lecture, students asked follow-up questions both in class and afterwards through email. In one case, a student explicitly noted that the interactive visualization supported their understanding of the conceptual differences between confined and unconfined aquifers.

7. Lessons learned and recommendations

- The implementation required little technical effort and could be integrated into an existing lecture with minimal preparation.
- Despite the low technical threshold, the interactive demonstration required additional classroom time, making it necessary to shorten or streamline other explanations.
- In this example, the subsequent theoretical discussion of storage and storage properties could be reduced, as key concepts had already been introduced effectively through the interactive visualization.
- The didactic approach is easily transferable to other courses and topics, particularly those addressing flow-to-well processes or pumping test interpretation.
- For first-time adopters, this example represents a concise and manageable entry point for gaining experience with interactive tools in classroom teaching.

Best Practice Example – Horton infiltration

1. Short Description of the Interactive Educational Tool

The interactive tool “Infiltration_Capacity_Intro” (available in the iNUX interactive materials and online under https://horton-infiltration-intro.streamlit.app/) visualizes how soil infiltration capacity evolves during a rainfall event according to the Horton infiltration model (Fig.1). The app also integrates initial and final multiple-choice assessments to support self-checking of conceptual understanding. The tool is designed primarily for live use in lectures and classroom settings, where immediate visual feedback helps students develop intuition for the link between rainfall, infiltration, and runoff generation.

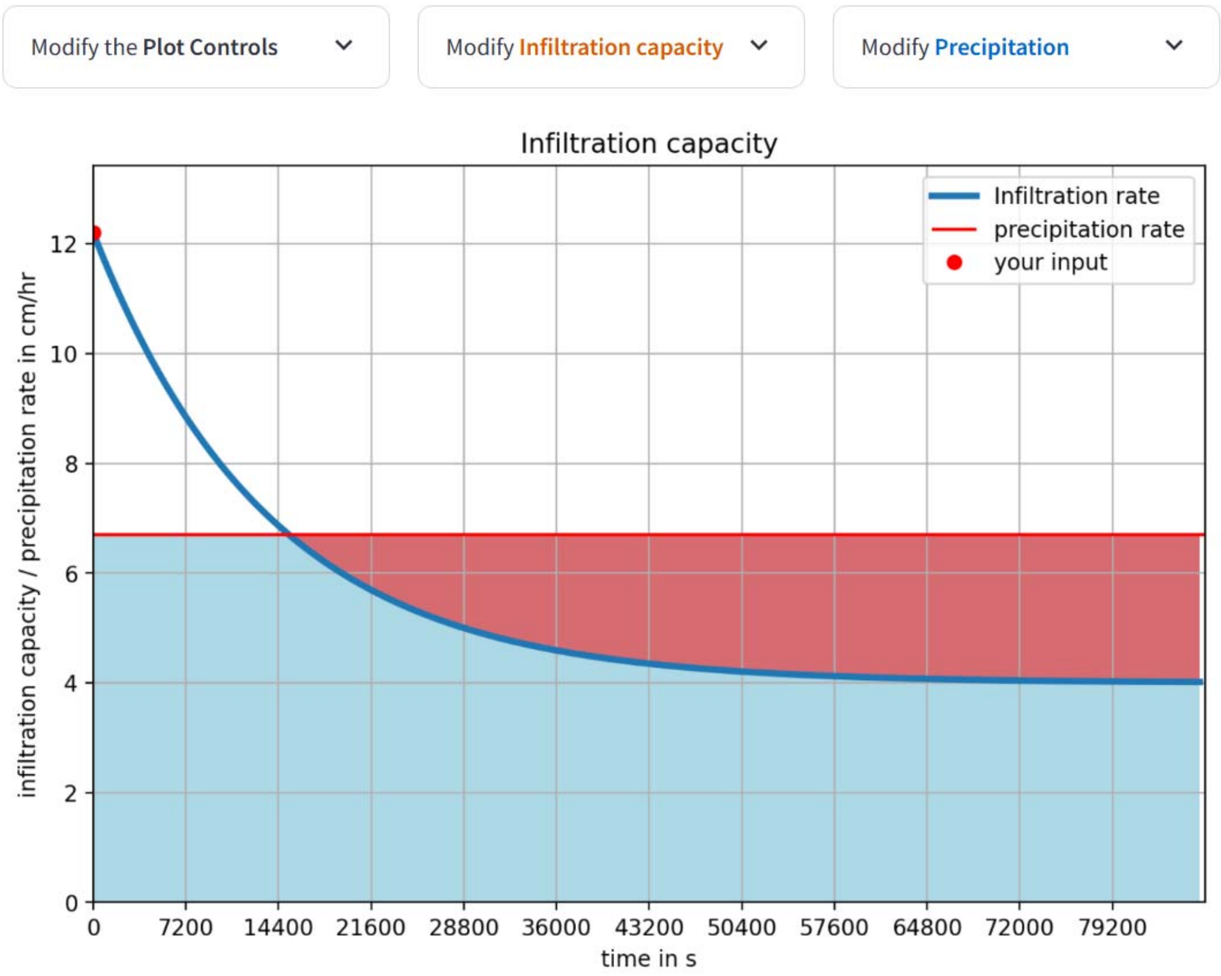

Figure 1: The interactive Streamlit app with the temporal distribution of infiltration and overland flow.

2. Context and educational setting

The tool was implemented in an introductory bachelor-level course in Hydrology and Hydrogeology at the University of Gothenburg, aimed at first-year students in Earth Sciences. The course focuses on fundamental hydrological processes and basic quantitative reasoning, with students typically lacking a strong background in mathematics or physics. The teaching approach therefore emphasizes conceptual, visually supported explanations rather than formal derivations.

The Horton infiltration app was used during a regular classroom lecture as a live, instructor-guided demonstration, embedded in a broader discussion of rainfall–runoff processes and streamflow dynamics. Around 20 students attended the session, which allowed for spontaneous questions and short discussions. The primary role of the app was to bridge theory and observation, linking the abstract Horton curve to the behavior of streamflow in a small, mountainous catchment with snow cover in winter.

3. Learning objective(s)

Conceptual understanding

- Explain the basic concept of infiltration capacity and how it typically changes over the course of a storm (high at the beginning, decreasing toward an equilibrium value).
- Describe the Horton infiltration model and the roles of the initial infiltration capacity f_0, the equilibrium capacity f_c, and decay rate k.

Process interpretation

- Relate the interaction between rainfall intensity and infiltration capacity to the onset of infiltration-excess (Hortonian) overland flow.
- Interpret how variations in infiltration capacity and rainfall forcing can contribute to features of a streamflow hydrograph in a small mountainous catchment (e.g. rapid peaks, low-flow periods, and seasonal signals).

Parameter sensitivity and comparison

- Explore how changing f_0, f_c, and k affects the timing and magnitude of overland flow generation and discuss how different parameter combinations might represent more or less permeable soils.
- Recognize the limitations of the Horton model as a **simple empirical tool** and distinguish it from more physically based descriptions of unsaturated flow.

4. Tool integration and didactic design

The interactive tool was introduced within a teaching unit on basic hydrological terminology, hydrological measurement tools, and fundamental equations and concepts. Before showing the app, students were presented with a streamflow time series from a small mountainous catchment with snow cover in winter. They were asked to interpret the hydrograph qualitatively and to discuss possible links between rainfall, snow processes, and streamflow behavior. At this stage, the class remained largely silent, and no clear explanations were given.

The Horton app was then introduced as a way to build intuition from first principles about infiltration and overland flow. The instructor briefly explained the conceptual meaning of f_0, f_c, k, and P, and demonstrated how the infiltration curve evolves over a synthetic storm. The app's plot shows both the infiltration capacity and the rainfall intensity, and highlights periods where rainfall exceeds capacity and thus generates potential overland flow.

Students were encouraged to follow along on their own devices, reading the explanation, adjusting parameters and observing how changes in soil properties and rainfall intensity affected the timing and duration of infiltration-excess conditions. This interactive phase lasted about 10–15 minutes. The embedded initial and final multiple-choice assessments framed the activity and supported reflection on key ideas, but the main emphasis in this implementation was on visual exploration rather than formal testing.

After working with the tool, the class revisited the original streamflow hydrograph. This time, several students actively contributed explanations, drawing connections between rainfall events and streamflow peaks, and asking questions about the long low-flow periods and pronounced spring peaks associated with snowmelt. The tool thus served as a concrete steppingstone from an abstract empirical model toward interpreting a real hydrological time series.

5. Implementation effort and technical setup (kept pragmatic)

The technical setup required for this implementation was modest. The Horton infiltration tool is deployed via Streamlit, accessible through a standard web browser without local installation or user accounts. In the classroom, the instructor ran the app live from a laptop connected to a projector, while students with laptops, tablets or phones could optionally access the same app directly via URL or provided QR code.

The initial and final assessment questions (Fig.2) were implemented using the streamlit_book multiple-choice component and JSON-based question files, which are easy to modify and reuse. Preparation mainly involved embedding the app link and QR code in the lecture slides, preparing the hydrograph example from the mountainous catchment, and ensuring that the sequence of explanation, interaction, and discussion fit within the available lecture time.

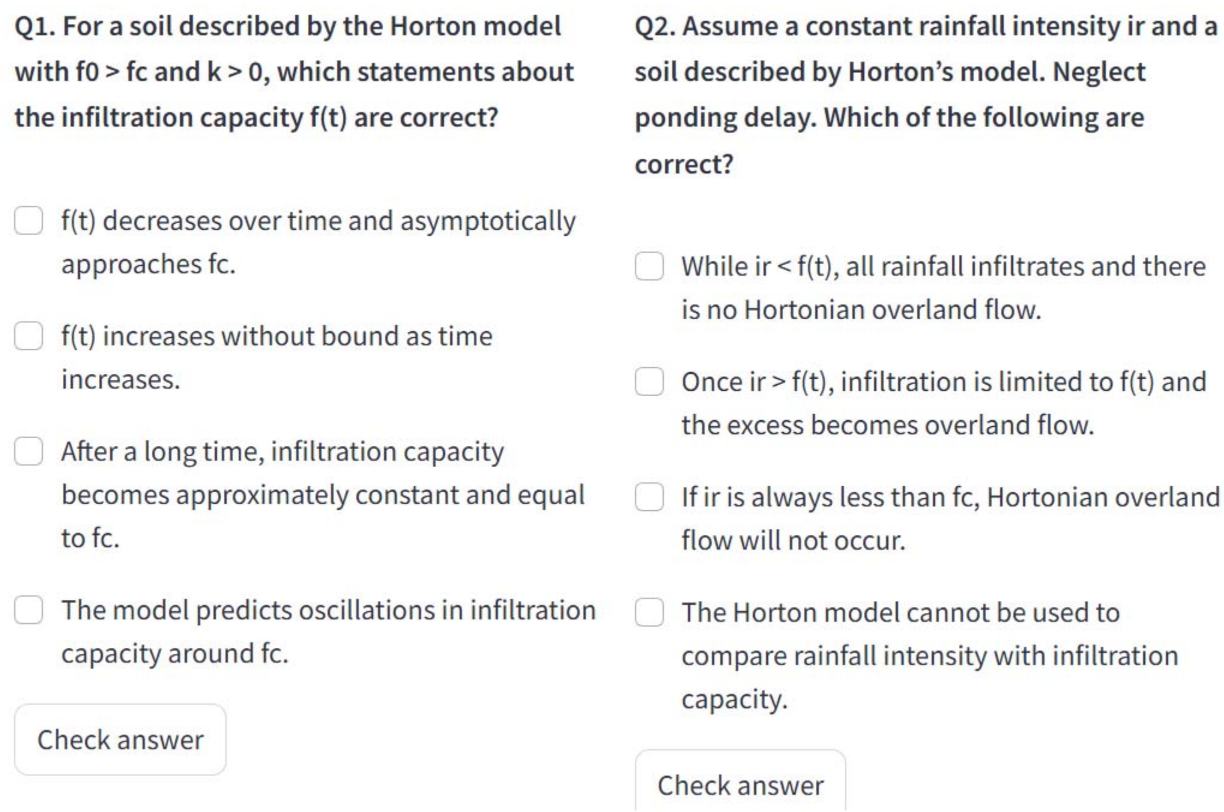

Figure 2: Example of final assessment questions at the end of the Streamlit app.

Once created, the tool is highly reusable in future iterations of the course and in other teaching contexts that address infiltration, runoff generation, or hydrograph interpretation.

6. Observed outcomes and feedback

During the session, students initially struggled to articulate explanations for the hydrograph features when only shown the streamflow plot. After working with the Horton infiltration app, at least a few students were able to link rainfall events to streamflow peaks and recognized that limited infiltration capacity can lead to rapid runoff responses. Some students went further and raised questions about the long low-flow periods, as well as the spring peaks likely driven by snowmelt, indicating that they were beginning to connect the simplified Horton concept with more complex seasonal dynamics in the catchment.

The interactive visualization appeared to lower the threshold for participation: after using the tool, students were more willing to share interpretations and ask questions.

7. Lessons learned and recommendations

- A simple empirical model like Horton's becomes very powerful when combined with interactive visualization and a real-world example (streamflow hydrograph).
- The tool required little extra technical effort and fit easily into an existing lecture, but did require dedicated classroom time, so other explanations had to be streamlined.
- Some theoretical content (e.g. long verbal explanations of infiltration capacity) could be shortened because the core ideas were already clear from the interactive exploration.
- The approach is easily transferable to other topics where infiltration-excess runoff matters (e.g. urban hydrology, floods, land-use change).
- For instructors new to interactive tools, this setup offers a manageable entry point to experiment with exploratory, concept-driven learning in hydrology.

Acknowledgment

This document was created within the ERASMUS+ project iNUX.

Co-funded by the European Union

This project is co-funded by the European Union. However, the views and opinions expressed are solely those of the author(s) and do not necessarily reflect those of the European Union or the National Agency DAAD. Neither the European Union nor the granting authority can be held responsible for them.

References

Reimann, T., Liedl, R., Giese, M., Barthel, R., Grießer, E., Birk, S., Bertran, O., Fernàndez-Garcia, D. (2025). Electronic Questions and Assessments in iNUX: Foundations, Design Principles, and Implementation Workflows. https://github.com/gw-inux/iNUX-Handbook/tree/main/Digital_Questions.

Reimann, T., , Giese, M., Barthel, R., Birk, S., Fernàndez-Garcia, D. (2025). Concept for Categorization of Hydrogeology and Groundwater Management Contents. https://github.com/gw-inux/iNUX-Handbook/tree/main/Categorization